

Heng Dong

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EXPERIENCE

- **ByteDance Seed-Robotics** Beijing, China
Embodied AI Researcher specializing in large-scale RL Training *Jun. 2025 - Present*
- **Tsinghua University** Beijing, China
Ph.D. majoring in Artificial Intelligence *Sep. 2020 - Jun. 2025*
- **University of Science and Technology of China (USTC)** Hefei, China
B.S. majoring in Computer Science and Technology *Sep. 2016 - Jun. 2020*

RESEARCH AREA

The goal of my research is to endow agents with superhuman intelligence, which I believe can be achieved through *learning from interactions* using *modern scalable models*. Toward this goal, my previous research mainly focused on

- **Learning from Interactions** RL, Robot Control & Design, Multi-Agent
- **Modern Scalable Models** Large Language Models, Diffusion Models, Flow Models

PUBLICATIONS AND PREPRINTS

Modern Scalable Models (Large Language Models, Diffusion Models, Flow Models)

- [1] Huang Fang*, Mengxi Zhang*, **Heng Dong***, Wei Li*, Zixuan Wang, Qifeng Zhang, Xueyun Tian, Yucheng Hu, Hang Li. “Robix: A Unified Model for Robot Interaction, Reasoning and Planning”. *Technical Report* (2025).
- [2] **Heng Dong***, Kefei Duan*, Chongjie Zhang. “Enhancing Decision-Making of Large Language Models via Actor-Critic”. *Forty-Second International Conference on Machine Learning (ICML 2025)*.
- [3] Tonghan Wang*, **Heng Dong***, Yanchen Jiang, David C. Parkes, Milind Tambe. “On Diffusion Models for Multi-Agent Partial Observability: Shared Attractors, Error Bounds, and Composite Flow”. *Proc. of the 24th International Conference on Autonomous Agents and Multiagent Systems (AAMAS 2025)*.
- [4] Zixuan Wang, Huang Fang, Shaoan Wang, Yuanfei Luo, **Heng Dong**, Wei Li, Yiming Gan. “Hydra-Nav: Object Navigation via Adaptive Dual-Process Reasoning”. *Forty-Third International Conference on Machine Learning (ICML 2026)*.
- [5] Xueyun Tian, Wei Li, Bingbing Xu, **Heng Dong**, Yuanzhuo Wang, Huawei Shen. “ROMA: Real-time Omni-Multimodal Assistant with Interactive Streaming Understanding”. *The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*.
- [6] Xueyun Tian*, Minghua Ma*, Bingbing Xu, Nuoyan Lyu, Wei Li, **Heng Dong**, Zheng Chu, Yuanzhuo Wang, Huawei Shen. “Learning from Mistakes: Negative Reasoning Samples Enhance Out-of-Domain Generalization”. *The 64th Annual Meeting of the Association for Computational Linguistics (ACL 2026)*.
- [7] Weichen Zhang, Ruiying Peng, Xin Zeng, Jianjie Fang, Ziyu Wang, kaiyuan Li, **Heng Dong**, Wei Li, Chen Gao, Xin Wang, Xinlei Chen, Yong Li. “The Point, the Vision and

the Text: Does Point Cloud Boost Spatial Reasoning of Large Language Models? A Bias-Controlled Study”. *Under Review* (2026).

- [8] Xinyi Yang, Liang Zeng, **Heng Dong**, Chao Yu, Xiaoran Wu, Huazhong Yang, Yu Wang, Milind Tambe, Tonghan Wang. “Policy-to-Language: Train LLMs to Explain Decisions with Flow-Matching Generated Rewards”. *The Fourteenth International Conference on Learning Representations* (ICLR 2026).

Learning from Interactions (Robot Design, Robot Control, Multi-Agent RL)

- [9] **Heng Dong***, Junyu Zhang*, Chongjie Zhang. “Leveraging Hyperbolic Embeddings for Coarse-to-Fine Robot Design”. In *The Twelfth International Conference on Learning Representations* (ICLR 2024).
- [10] **Heng Dong**, Junyu Zhang, Tonghan Wang, Chongjie Zhang. “Symmetry-Aware Robot Design with Structured Subgroups”. In *Fortieth International Conference on Machine Learning* (ICML 2023).
- [11] **Heng Dong**, Tonghan Wang, Jiayuan Liu, Chongjie Zhang. “Low-Rank Modular Reinforcement Learning via Muscle Synergy”. In *Thirty-sixth Conference on Neural Information Processing Systems* (NeurIPS 2022).
- [12] **Heng Dong***, Tonghan Wang*, Jiayuan Liu, Chi Han, Chongjie Zhang. “Birds of a Feather Flock Together: A Close Look at Cooperation Emergence via Multi-Agent RL.” *arXiv preprint* (2021).
- [13] Yihan Wang*, Beining Han*, Tonghan Wang*, **Heng Dong**, Chongjie Zhang. “DOP: Off-Policy Multi-Agent Decomposed Policy Gradients”. In *Ninth International Conference on Learning Representations* (ICLR 2021).
- [14] Tonghan Wang, **Heng Dong**, Victor Lesser, Chongjie Zhang. “ROMA: Multi-Agent Reinforcement Learning with Emergent Roles”. In *Thirty-seventh International Conference on Machine Learning* (ICML 2020).

HONORS AND AWARDS

• ByteDance Seed - SpotBonus Award	<i>Jan. 2026</i>
• Tsinghua Friends - Ubiquant Excellence Scholarship	<i>Sep. 2024</i>
• Interdisciplinary Information Institute Scholarship	<i>Sep. 2023, Sep. 2022</i>
• Huiyan Scholarship of Excellence	<i>Sep. 2021</i>
• Outstanding Undergraduate Thesis Award	<i>Jun. 2020</i>
• Scholarship for HUA Xia Talent Program (top 30)	<i>Aug. 2017 - Jul. 2020</i>
• Scholarship for Excellent student	<i>Oct. 2016, Oct. 2017, Oct. 2018</i>

RESEARCH EXPERIENCE

Large Foundation Models for Robotics

Researcher

ByteDance Seed-Robotics, Beijing, China

Jun. 2025 - Present

- Leader: Wei Li, Hang Li and Chang Zhou
- Build Embodied Intelligence and Robotic Foundation Models
- Large-Scale Reinforcement Learning for Embodied Reasoning and Interaction in IsaacSim

Modern Scalable Models: Diffusion Models and Flow Models

Cooperation

Aug. 2024 - May 2025

- Supervisor: Prof. Milind Tambe and Prof. David C. Parke
- Diffusion Models, Rectified Flow for Explainable LLMs

Learning from Interactions & Modern Scalable ModelsTsinghua University, Beijing, China

Ph.D. Student

Sep. 2020 - Jun. 2025

- Supervisor: Prof. Chongjie Zhang and Prof. Yi Wu
- Reinforcement Learning, Large Language Model, Robot Design, Multi-Agent

Multi-Agent: Role-Based, Self-Interested

Tsinghua University, Beijing, China

Intern

Sep. 2019 - Jul. 2020

- Supervisor: Prof. Chongjie Zhang
- Role-Oriented Multi-Agent Systems, Self-Interested Agents

Knowledge Graph of Intelligent Healthcare

USTC, Hefei, China

Lab Research Work

Sep. 2018 - Jun. 2019

- Supervisor: Prof. Tong Xu
- Intelligent Healthcare based on Knowledge Graph from electronic medical records

PROFESSIONAL SERVICES

Reviewer

- Association for Computational Linguistics (ACL) *2026 - Present*
- Annual Conference on Neural Information Processing Systems (NeurIPS) *2022 - Present*
- International Conference on Machine Learning (ICML) *2022 - Present*
- International Conference on Learning Representations (ICLR) *2022 - Present*
- Association for the Advancement of Artificial Intelligence (AAAI) *2025 - Present*

Teaching Assistant

- Artificial Intelligence: Principles and Techniques (Tsinghua, IIIS) *Fall, 2021*
- Reinforcement Learning (Tsinghua, IIIS) *Spring, 2022*

ENGINEERING SKILLS

- **Frameworks** PyTorch, Verl, Transformers, Numpy, Matplotlib, Plotly, Git
- **Programming Languages** Python, C, Wolfram
- **OS** Linux (Ubuntu, Deepin, OpenSUSE), MacOS, Windows